

**A**  
**Report**  
for the course:

***Financial systems (BM502H)***



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## **1. Introduction:**

When India was still under the control of British, the bank of Calcutta was established in 1806. It was the first bank of India which used to fund the trade of the British East India Company in Bengal. Along with Bank of Bengal (renamed after three years), Bank of Bombay (1840) and Bank of Madras (1843) was established. Later, in 1921 it was named as Imperial Bank of India. In 1955 the Imperial Bank of India was nationalized and was renamed SBI (in 1955) with RBI (Reserve Bank of India) acquiring 60% of stake. In the era from 1959 to 2017, SBI and its associates were formed which included state bank of Bikaner and Jaipur, State Bank of Mysore, and State Bank of Patiala. The SBI currently is the largest public sector bank in India.

In the time duration between 1969-1991 fourteen major banks were nationalized with the deposit more than rupees 50 crore which includes (Allahabad bank, Bank of India, Bank of Baroda, Bank of Maharashtra, Central Bank of India, Canara Bank, Dena Bank, India Overseas Bank, Indian Bank, Punjab National Bank, Syndicate Bank, Union Bank of India, United Bank, UCO bank). Further in 1980 another six banks (Andhra Bank, Corporation Bank, New Bank of India, Oriental Bank of Commerce, Punjab and Sindh Bank, Vijaya Bank). The efficiency of these banks was not benchmarked until they were nationalized but it was not the sole reason to nationalize these banks. There were other reasons like to increase funds in the bank which will further increase the economy of the country. Also, it boosted the rural and agricultural sectors along with increasing employment opportunities in the country. Even after having number of reasons to nationalize banks, the major priority of the government was to reduce the number of the small banks and increase the number of large banks like SBI in India at a global level.

This report highlights the analysis of liquidity creation and financial ratios of State Bank of India and Yes Bank from 2006 to 2020. The analysis will include a comparison of liquidity creation and financial ratios on a yearly basis as well as a comparison between both the banks. We will be observing the yearly trends with the data which we have calculated. The balance sheet of the banks includes assets and liabilities with financial year as of 31<sup>st</sup> March of every year. The assets and liabilities will be used in calculating liquidity creation. There are other aspects of the balance sheet which will be used to calculate different financial ratios required for the project. The balance sheet acknowledges the banks financial status.

## 1.1 Liabilities:

Alexander Hamilton Church, an early 20th-century management and accounting theorist defined liabilities as "Obligations or debts that a business must settle in the future, arising from past transactions or events. These obligations represent claims by creditors against the assets of the business, reflecting the company's commitment to provide economic benefits to others." Another pair of profound economists named Paul A. Samuelson and William D. Nordhaus quoted "Liabilities are debts or obligations that arise during the course of business operations. They represent what the company owes to others and must pay in the future, including loans, accounts payable, and other financial commitments." In simple words, liabilities refer to the financial debt or obligations that a company or a bank has to repay to its creditors, lenders, or customers which could be in the form of cash or assets.

Liabilities are of three types which includes:

- Current liabilities
- Non-current liabilities
- Contingent liabilities

Current liabilities are those liabilities which are for a short period of time and requires payment on regular intervals nearly within a year for a company or bank. For example, salaries, rents, bills payable, accounts payable, short-term loans and borrowings, dividends payable, taxes, etc. Non-current liabilities are used to finance company's long-term investments, capital expenditures, and other strategic initiatives. These entities are not bound for repayment within 12 months. For example, long term borrowings, subordinated debt, deferred tax liabilities, pension and retirement obligations, leases liabilities, etc. Contingent liabilities are a type of provisions kept by banks or companies for future uncertain events.

Another classification of liabilities is done based on its liquidity which includes liquid liabilities, semiliquid liabilities, and illiquid liabilities. Liquid liabilities are those entity which can be easily converted to cash within a short span of time whereas semi-liquid liability can also be converted to cash in a short time, but it requires longer steps to settle. Illiquid liability on the other hand are entities which cannot be converted to cash easily or in a short duration. Usually, it requires more than a year to convert into a cash. The weightage for liquid liabilities is 1/2, semi-liquid liability is 0, and illiquid liability is -1/2.

| <b>Liquid liability (1/2)</b> | <b>Semi-liquid liability (0)</b> | <b>Illiquid (-1/2)</b>                          |
|-------------------------------|----------------------------------|-------------------------------------------------|
| Demand deposits               | Term Deposits                    | Subordinate debt                                |
| Saving deposits               | Borrowings from RBI              | Perpetual debt                                  |
| Bills Payable                 | Borrowing from other banks       | Borrowings from other institutions and agencies |
| Inter-office adjustments      |                                  | Deferred tax liabilities                        |
| Interest accrued              |                                  | Capital                                         |
|                               |                                  | Reserves and surplus                            |
|                               |                                  | Others (including provisions)                   |

*Table 1 Classification of liabilities into liquid, semi liquid and illiquid liabilities along with its weightage*

## **1.2 Assets:**

Irwing Fisher quotes "Assets are goods or resources owned by an individual or a business that have economic value and can generate future income or benefits." Later, Paul A. Samuelson defined assets as economic resources that are owned by an entity, which have the potential to produce value either through use or by generating cash flows. In other words, assets are defined as resources owned by individuals, companies, and banks that are expected to provide future benefits in the form of cash or other equally valuable entity. The assets can be classified into:

- Current Assets
- Non- current assets
- Tangible vs Intangible assets
- Operating vs non-operating assets

Current assets can be converted into cash easily which can be consumed withing a year. The banks or companies can easily convert current assets under a short span of time i.e. one year. Examples of current assets include cash and cash equivalents, account receivables, inventory, and prepaid expenses. Non- current assets can be defined as the assets which are fixed or are for a longer period which cannot be converted to cash easily. Examples include plant and machinery, long term investments, deferred tax assets, etc. Tangible assets are a physical assets like land, machinery, and buildings while Intangible assets are non-physical assets such as intellectual property rights such as patents and brand recognition. On the other hand, operating assets are those assets which are used in companies' daily operations like inventory

and machines. Non-operating assets cover core operations for the banks or companies like excess cash and marketable securities. The other class of assets include liquid assets, semiliquid assets, and illiquid assets. Liquid assets are the entities which can easily be converted to cash with low transaction costs while illiquid assets are those assets which take a longer period to get converted into cash, but it has a high transaction cost. Semiliquid assets on the other hand are more liquid than illiquid assets but less liquid than liquid assets with a moderate transaction cost.

| <b>Liquid Asset (-1/2)</b>              | <b>Semi-liquid Asset (0)</b>     | <b>Illiquid Asset (1/2)</b>                   |
|-----------------------------------------|----------------------------------|-----------------------------------------------|
| Cash in hand                            | Balances with bank in India      | Subsidiaries and joint ventures in India      |
| Balances with RBI                       | Balances with bank outside India | Subsidiaries and joint ventures outside India |
| Money at Call and short notice          | Advances                         | Fixed Asset                                   |
| Government securities in India          |                                  | Other Asset                                   |
| Other approved securities               |                                  |                                               |
| Shares                                  |                                  |                                               |
| Debentures and Bonds                    |                                  |                                               |
| Other investment inside + outside India |                                  |                                               |
| Government securities outside India     |                                  |                                               |

*Table 2 Classification of assets into liquid, semi liquid and illiquid liabilities along with its weightage*

### **1.3 Liquidity Creation (SBI):**

Liquidity creation is a process in which financial institutions like primarily banks easily convert their liquid assets into cash. This process is essential for the government to ensure that money is rightly funded for the sectors which lead to economic growth of the country. SBI is the largest public sector bank of India, and it is necessary to understand the trends of liquidity created by SBI over the years for future prediction of economic growth of the country. In our project, we have taken the data of liquidity creation from the year 2006 to



2020. The project focuses on key elements of liquidity creation such as liquid assets, cash reserves, balances with RBI, short term money market, and investments in government securities.

There are following observation we found during the analysis:

**Cash in Hand:**

SBI's has always adopted a secure approach towards the banking sector and so their efforts were to maintain ready cash for immediate needs and potential crises. This can be seen in our analysis where there is a steady growth from 2006 to 2020, reflecting a strengthening of liquid cash reserves. In 2006, the cash in hand was 0.90 crores which steadily increased to 1630.74 crores in the year 2020.

**Balances with RBI:**

Balance with RBI ensures solvency and compliance with regulatory requirements. We have observed a substantial reserve was maintained consistently by SBI, with a peak around 2018. The RBI had a balance of 10803.8 crores of reserve in the year 2018.

**Money at Call and short notice:**

This component showed fluctuations, with notable dips in certain years (e.g., 2016,2018), indicating a shift in short term lending or borrowing strategies.

**Government securities in India:**

SBI, as we know, always had a secure approach and so SBI has significant investment in government securities which reflects its low risk and long-term investment strategy. Investment has peaked in certain years like 2020 according to our analysis.

**Capital:**

The base capital of SBI has remained stable, with minor changes over the years.

**Reserves and Surplus:**

SBI has increased its financial strength over time with steady growth in reserves and surplus which includes statutory reserves, capital reserves, and share premium. Statutory reserves have gradually increased from 43,810 crores in 2014 to 69,942 crores in 2020, reflecting SBI's commitment to maintaining robust liquidity buffers.

### **Patterns Observed:**

Prior to the global financial crises of 2008, the growth of liquidity component was moderate. Following the financial crises, there were discernible rises in cash reserves and government securities, due to SBI's cautious strategy of ensuring liquidity.

Impact of demonetization (2016) reflected as a sudden flood of liquidity in SBI. During this time, the high value currency notes were adjusted to cash reserves and balances with the RBI.

Following 2019, there was a notable surge in all liquidity measures, most likely because of banks increasing their reserves to manage possible shocks and preserve solvency in the wake of the covid-19 pandemic.

Over the period, SBI's approach to liquidity creation has significantly changed, with a major focus on preserving the harmony between short term securities and long-term assets.

### **1.4 Liquidity Creation (YES Bank):**

As we already know the factors affecting the liquidity creation of the bank, we will try to understand directly the trend of YES bank. We have observed a negative liquidity creation over the years in YES highlighting its inability to create liquidity. The dynamic shortfall suggests systemic challenges in asset management and risk exposure. The rapid growth in liquid liabilities has resulted in the inability of the bank to increase its liquid assets. This imbalance places significant strain on the bank's liquidity position and operational stability. The sharp drop in liquid assets in 2020 may be indicative of the bank's challenges in accessing or retaining liquid funds, possibly influenced by market conditions or operational decisions. We have also observed the reduction in liabilities with positive trend, but it is insufficient to offset the liquidity crisis. The persistent liquidity issues could have regulatory implications, potentially affecting the bank's operational capabilities and customer confidence.

### **1.5 SBI vs YES Bank:**

**SBI Dominance:** SBI, the biggest public sector bank in India, consistently grows and demonstrates a far superior ability to create liquidity, particularly after 2016. This illustrates the bank's capacity to raise deposits and make large-scale loan extensions.

**YES, Bank's Challenges:** Conversely, Yes Bank exhibits a steep fall in liquidity creation after 2018, indicating stagnation in the sector. This illustrates its liquidity problems, which are

probably connected to its problems with asset quality and governance. The fall of YES bank started in the year 2018 but YES bank failed drastically in the year 2020 due to the reasons including higher loans granted than deposits (2.25 trillion loan and 2.10 trillion deposit), NPAs (non-performing assets) reached to 17,134 crores in the year 2019, NBFCs crises (total exposure to IL & FS and DHFL was 11.5 percent in 2019), failed governance compliance, and at last the excessive withdrawals due to the financial review outreach to the public. After the failure of YES bank, RBI took over the YES bank management and imposed a moratorium on the lender. RBI announced a draft ‘Scheme of Reconstruction’ that allowed SBI investing capital to acquire 49% stake in the restructured private lender.

Divergence: By 2020, SBI and Yes Bank's approaches to creating liquidity will diverge significantly. While Yes Bank's liquidity situation is getting worse, SBI's ability to create liquidity is still growing.

In conclusion, SBI established itself as a powerful liquidity generator in the market by demonstrating notable growth and resilience in its liquidity creation from 2006 to 2020.

|                    | 2020        | 2019       | 2018       | 2017   | 2016   | 2015    | 2014    | 2013    | 2012    | 2011    | 2010    | 2009    | 2008      | 2007      | 2006      |
|--------------------|-------------|------------|------------|--------|--------|---------|---------|---------|---------|---------|---------|---------|-----------|-----------|-----------|
| Liquidity SBI      | 1587980.017 | 1406348.5  | 1242243.66 | 883307 | 778390 | 719041  | 622085  | 567212  | 489488  | 436782  | 354081  | 295139  | 234300.32 | 201849.82 | 170541.34 |
| Liquidity Yes Bank | -53167.8107 | 37350.9149 | 47944.2982 | 37470  | 25271  | 17145.4 | 12384.5 | 8007.82 | 5780.44 | 10038.4 | 4998.32 | 2745.75 | 2687.6046 | 1339.8749 | 492.15585 |

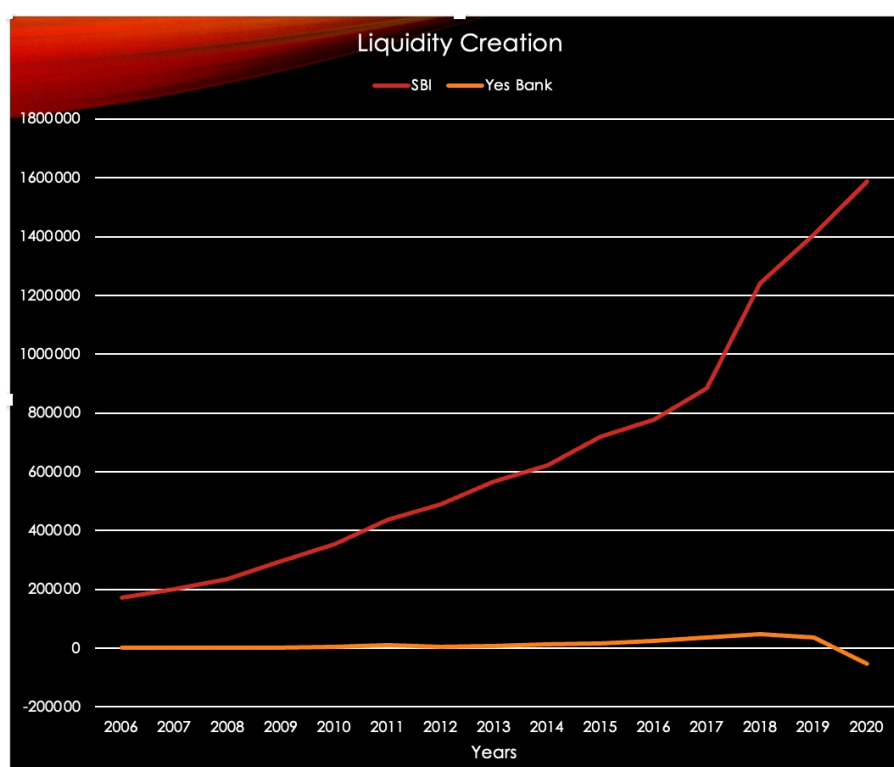


Figure 2 Comparative analysis of liquidity creation of YES bank and SBI over the years

## 2. Financial Ratios:

### 2.1 Operating Efficiency:

It is a measure of efficiency of a bank managing its operational cost in compare with the revenue the bank is generating. In terms of mathematical calculations, it is expressed as operating expenses divided by total revenue which is then multiplied by 100. A lower operating efficiency ratio indicates better efficiency, which means bank can generate more income with fewer expenses.

Operating expenses includes all the costs related to day-to-day operations of the bank, such as salaries, rent, utilities, administrative costs, and other general expenses. The total amount of income earned by the bank from its operations including interest incomes, services fees, revenue, etc. is call the total revenue of the bank. With our analysis, we have analysed operating efficiency of SBI which revealed in 2006, SBI had 2.3 % OER which declined to 1.9 % in the year 2020. For YES bank, it was 2.6 % in the year 2006 and it remained same till the year 2020 that is 2.6 %.

**Operating Efficiency Ratio = (Operating Expense/ Total Revenue) \* 100**

### 2.2 Cost of funds:

It refers to the interest rate a institution or bank pays on its borrowings or deposits. It is a direct indicator of banks' profit and liquidity creation. Lower cost of funds enables banks to achieve a higher **Net Interest Margin (NIM)**, the difference between the earnings from loans and the payment for deposits and borrowings. A lower cost of funds allows banks to offer loans at competitive rates without fluctuating profitability. It affects a bank's ability to maintain liquidity while balancing the cost of acquiring that liquidity. Banks with higher costs of funds might need to adjust their loan pricing or reduce reliance on high-cost borrowings to remain competitive. The cost of funds for SBI in 2006 was 4.9% whereas for the year 2020 it was 4.4%. Moving towards YES bank, we found that cost of funds was 3.1% in 2006, and it increased up to 8.7% in the year 2020.

**Cost of funds = (Interest Expenses / Total deposits + Total borrowing) \* 100**

### 2.3 Income productive per employee:

This ratio measures the efficiency of workforce of the company or the bank in generating revenue for the banks it measures how much of the income is generated for each employee within the firm or the bank income productivity per employee ratio is essential to understand the overall contribution of human resource to their performance ratio. In the year 2006, YES bank has an income productive per employee ratio of 3.5% and for the year 2020, it increased to 51%. The SBI had 0.2% in the year 2006 which slightly increased to 1.2% in 2020. This data reveals the significant difference in the employee productivity in a government sector and the private sector, and the rate at which it is growing.

**Income productive per employee = (Total income / Full time employees) \* 100**

### 2.4 Efficiency ratio:

It measures the bank's efficiency in terms of the expenses banks are having for operations compared to its total income. The less the efficiency ratio of the bank, the more is the efficiency of the bank, that means the bank is using less of its income to cover its operating cost. To improve efficiency, banks try to increase their revenue. The efficiency ratio of less than 50% is excellent, which means banks are doing well in terms of efficiency. The efficiency ratio of SBI for the year 2020 was 0.8%, which was the same in the year 2006. YES bank on the other hand had an efficiency ratio of 1.7% in 2006 which later changed to 67.2% in the year 2020. This indicates that after downfall, the efficiency ratio of YES bank also declined.

**Efficiency ratio = (Total operating expense + Provisions) / (Net interest income + Other total income)**

### 2.5 Demand deposit ratio:

Demand deposit ratio reveals the funding of bank assets. It gives the value of the deposit money that is used to fund the bank's assets. Deposits such as current accounts, are typically low or no interest-bearing. A higher ratio means that the bank is using cheaper funding sources to support its asset base, which can reduce the **cost of funds** and improve profitability. If a large proportion of the bank's assets are funded by low-cost demand deposits, it can lend at a higher margin. This boosts the **Net Interest Margin (NIM)**, positively impacting profitability. The analysis revealed that SBI had DD ratio 13.76% in the

year 2006 which declined to 5.7% in the year 2020. YES bank had 7.2% DD ratio in the year 2006 which further decreased to 3.6% in the year 2020.

**Demand deposit ratio = (Total demand deposit / Total asset) \* 100**

## **2.6 SLR investment-Total investment ratio:**

SLR investment to total investment ratio shows how much the bank has invested in SLR and how much the bank is following its regulatory compliances. It also indicates banks conservative investment strategy as SLR is a safe investment, but it yields low returns. Now, banks need to manage or balance between maintaining regulatory compliance and profitable investments. The analysis shows the given ratio of SBI was 0.7% in the year 2006 which declined to 0.6% in the year 2020. On the other hand, YES bank's SLR to total investment ratio in the year 2006 was 91.4%, which in future declined to 16.1% in the year 2020. Thus, YES bank has shown a shift from conservative investment to high return yielding investments with a higher risk.

**SLR-Total investment ratio = SLR \* Total Asset/ Investments**

## **2.7 Cash to demand deposit ratio:**

The cash to demand deposit ratio includes cash in hand, balances with RBI, balances with banks in India, money at call and short notice, and total deposits. Demand deposit ratio is the liability for the bank which indicates the ratio between the liquid asset to the liability. Increase in the cash to demand deposit ratio indicates that the bank has enough liquid cash to cover its immediate demand deposit, it helps to handle immediate large withdrawals preventing them from bank run or selling their assets. The cash to demand ratio with 0.5% or above is considered good whereas below 0.2 is poor. The banks which keep the cash to DD ratio very high implies that it has lost its opportunity costs and the banks with low cash to DD ratio reveal that the bank is not meeting its obligations. In 2006, SBI had a cash to DD ratio of 0.5% whereas in 2020, it was 0.9%. For YES bank, the analysis revealed 0.6% in 2006, which slightly increased to 0.8% in 2020.

**Cash to demand deposit ratio = (Cash in hand + Balances with RBI + Balances with banks in India + Money at call and short notice) / Demand deposit**

## **2.8 Cash to Total deposit ratio:**

The cash to total deposit ratio includes cash in hand, balances with RBI, balances with banks in India, money at call and short notice, and total deposits. Increases in the cash to total deposit ratio indicate high liquidity and the bank's ability to handle large withdrawals. The ratio above 0.2% of the banks indicates its high liquidity management and the ratio lower than that reveals the bank's potential liquidity risk. The difference between the cash to total deposit ratio and cash to dd ratio of the banks reveals the ability of the bank to handle large withdrawal of short-term deposits and large withdrawal of long-term deposits. The cash to total deposit ratio for SBI was 10.36% in 2006, which declined to 6.83% in the year 2020. YES bank had 7.03% ratio in 2006, and 7.5% in 2020.

**Cash to total deposit ratio = (Cash in hand + Balances with RBI + Balances with other banks in India) / Total deposits**

## **2.9 Equity Multiplier:**

It is the financial metric that measures the proportion of banks' assets that are financed by its shareholders' equity. It shows how much the bank's assets are financed by equity vs debt. Increase in equity multiplier shows higher financial leverage which amplifies returns but also increases financial risks. Decrease in equity multiplier means banks rely on more equity than debt, indicating lower financial leverage, and typically lower financial risks. The equity multiplier for SBI in the year 2006 and 2020 was 17%, and that of YES bank was 7.2% in 2006, which further increased to 11.8% in 2020.

**Equity Multiplier = Total asset / Equity (Reserves and surplus + capital)**

## **2.10 Equity ratio:**

This ratio shows how much of the bank's assets are funded by equity rather than debt. An equity ratio closer to 1 indicates that a larger portion of the bank's assets is funded by equity, implying lower financial risk and vice versa. The equity ratio of SBI in the year 2006 is 0.05% and the same in the year 2020 as well. YES bank had an equity ratio of 0.13% in 2006, which slightly declined to 0.08% in the year 2020.

**Equity ratio = Equity / Total asset**

### **2.11 Capital adequacy ratio:**

Capital adequacy ratio indicates the bank's ability to withstand loss without affecting operating efficiency. In other words, it shows the total capital the bank has in reserve to cover its risk and protect depositors against losses. SBI had a capital adequacy ratio of 13.02% in the year 2006 and 14.9% in the year 2020. On the other hand, YES bank was having 26.5% capital adequacy ratio in 2006, which increased to 54.52% in the year 2020.

**Capital adequacy ratio = (Total capital + reserves and surplus + total borrowings + investments) / (Advances + total assets)**

### **2.12 Provision ratio:**

Provision ratio is a financial metric which determines the bank's ability to manage the credit risk. In simple language, the banks' ability to withstand a loss done by non-performing assets. SBI had a provision ratio of 57.5% in the year 2006 which increased to 66.17% in the year 2020. YES bank had a provision ratio of 61.5% in the year 2006 and 79.3% in the year 2020.

**Provision ratio = Advances / Total assets**

### **2.13 Net NPA to asset:**

Net NPA to asset measures the bank's assets that are non-performing. Increase in the NPA to total asset ratio increases the NPA and increases the credit risk of the bank indicating a potential financial stress for the bank. The net NPA to asset ratio for SBI was 0.7% in the year 2006 and 2.4% in the year 2020. That of YES bank continuously declining from 2006 and was negative until 2018, but in 2020 it became positive and was 1.7%.

**Net NPA to asset = (Gross NPA + Provisions) / Total assets**

### **2.14 Net NPA to equity:**

This ratio reveals banks equity that has turned into NPA. Increase in the net NPA to equity ratio implies an increase in higher banks equity is tied up with NPA. This reveals that a significant part of equity may be exposed to losses from bad loans. The net NPA to equity ratio of SBI was 12.6% in 2006 which increased to 41.1% in 2020. For YES bank, it was negative initially from 2006, but in 2020 it went to 20.83%.

**Net NPA to equity = (Gross NPA – Provisions) / Equity**



## **2.15 Return On Equity:**

Return on equity suggests how well a company is using its equity shareholders' equity to generate profit. It includes profitability, operational efficiency, and financial leverage. The return on equity for SBI is 15.9% in the year 2006, and 6.2% for the year 2020. The return on equity for YES bank for the year 2006 is 9.6% and in the year 2020 is -75.5%. This is due to the collapse of YES bank in the year 2020. The shareholders suffered a major loss.

$$\text{ROE} = \text{Net profit loss during the year} / \text{Equity}$$

## **2.16 Return on Asset:**

This ratio implies how well the bank uses its financial assets for generating profits. It also indicates the bank's efficiency in converting its investments into asset, and then into earnings. The ROA for SBI for the year 2006 was 0.89% and in 2020 it was 0.36%. On the other hand, the ROA for YES bank in the year 2006 was 1.3% whereas that in the year 2020 was -6.3%.

$$\text{ROA} = \text{Net profit} / \text{Total asset}$$

## **2.17 Profit Margin:**

This ratio evaluates a bank's profitability and efficiency in generating earnings from their operations. The profit margin for SBI for the year 2006 was 10.1% whereas that in the year 2020 was 4.7%. YES bank had a profit margin of 56.9% in the year 2006, which drastically declined to -138.4% in the year 2020.

$$\text{Profit Margin} = \text{Net profit loss during the year} / \text{Total income}$$

## **2.18 Asset utilization ratio:**

Asset utilization indicates how well the bank is using its assets to generate revenue and profit. It also gives insights into operating efficiency, the financial health of the bank, and the overall asset management strategies of the bank. SBI in the year 2006 had asset utilization of 8.7% and in the year 2020 it was 7.6%. YES bank had asset utilization of 2.3 in 2006 and in the year 2020, it was 4.5%.

$$\text{Asset utilization} = \text{Total income} / \text{Total asset}$$

### **2.19 Interest cost to asset ratio:**

Interest cost to asset ratio indicates how much a bank's asset is allocated or used to cover interest expenses. This ratio also helps in evaluating the effectiveness of asset and liabilities management strategies of the bank. The interest cost to asset ratio for SBI in the year 2006 was 4.12% and in the year 2020, it was 4.02%. For YES bank, analysis revealed this value to be 2.5% in 2006 and 7.4% in 2020.

**Interest cost to asset ratio = Total interest expended / Total assets**

### **2.20 Net operating margin:**

Net operating margin indicates the profitability of the bank based on its operating activities. This ratio also implies how well a bank is converted its revenues into profit after covering its expenses. The core idea of this ratio is the earnings generated from the operations of the bank. SBI had a net operating margin of -0.86% in the year 2006 and in 2020, it slightly increased to -0.75%. YES bank had -2.06% of net operating margin in 2006 and in 2020 it went up to 1.9%.

**Net operating margin = (Total other income + total operating expense) / Total assets**

### **2.21 Net interest margin:**

This ratio measures the difference between interest income generated from loans and other interest earning assets, and the interest paid on deposits, and other interest-bearing liabilities. The net interest margin of SBI in the year 2006 was 3.15% and in 2020 it was 2.4%. YES bank on the other side had 2.1% net interest margin in the year 2006, and 2.6% in 2020.

**Net interest margin = (Total interest income – total interest expense) / Total assets**

**Classification of ratio based on key performance indicators (KPI):**

| <b>Profitability</b>    | <b>Risk</b>       | <b>Liquidity</b>             | <b>Efficiency and Expense Control</b> |
|-------------------------|-------------------|------------------------------|---------------------------------------|
| ROE                     | Equity Multiplier | Demand Deposit               | Operating Efficiency                  |
| ROA                     | Equity Ratio      | SLR-Total Investment         | Cost of Funds                         |
| Profit Margin           | Capital Adequacy  | Cash to Demand Deposit Ratio | Income Productive per Employee        |
| Asset Utilization       | Provision Ratio   | Cash to Total Deposit Ratio  | Efficiency Ratio                      |
| Interest Cost to Assets | Net NPA to Asset  |                              |                                       |
| Net Operating Margin    | Net NPA to Equity |                              |                                       |
| Net Interest Margin     |                   |                              |                                       |

*Figure 3 Key Performance Indicator*

### 3. Key Performance Indicators and its analysis via graphical representation:

#### 4. State Bank of India:

##### 4.1 Profitability:

| Profitability           | 2020     | 2019     | 2018     | 2017     | 2016     | 2015     | 2014     | 2013     | 2012     | 2011     | 2010     | 2009     | 2008     | 2007     | 2006     |
|-------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| ROE                     | 6.244675 | 0.390301 | -2.98795 | 5.568178 | 6.897032 | 10.20068 | 9.207782 | 14.26422 | 13.94535 | 12.71738 | 13.89866 | 15.74045 | 13.72376 | 14.50964 | 15.94074 |
| ROA                     | 0.366658 | 0.023424 | -0.18952 | 0.387444 | 0.422064 | 0.6397   | 0.607513 | 0.90058  | 0.87661  | 0.675351 | 0.870128 | 0.945761 | 0.932624 | 0.801551 | 0.891987 |
| Profit Margin           | 4.788745 | 0.308332 | -2.46981 | 4.96926  | 5.186855 | 7.487769 | 7.03093  | 10.39486 | 9.685619 | 8.500934 | 10.6629  | 11.92641 | 11.67334 | 10.31937 | 10.15017 |
| Asset Utilization       | 7.656667 | 7.597122 | 7.673489 | 7.796814 | 8.137184 | 8.543269 | 8.640573 | 8.663706 | 9.05063  | 7.944438 | 8.160333 | 7.929975 | 7.989346 | 7.767435 | 8.7879   |
| Interest Cost to Assets | 4.029939 | 4.197864 | 4.215805 | 4.200293 | 4.530145 | 4.754787 | 4.856713 | 4.809428 | 4.734516 | 3.993341 | 4.492297 | 4.449799 | 4.425213 | 3.915548 | 4.127379 |
| Net Operating Margin    | -0.75802 | -0.89415 | -0.44411 | -0.40695 | -0.59115 | -0.75573 | -0.95791 | -0.84584 | -0.87738 | -0.58761 | -0.50792 | -0.3067  | -0.54242 | -0.89279 | -0.86835 |
| Net Interest Margin     | 2.482284 | 2.400188 | 2.166689 | 2.28605  | 2.425958 | 2.686187 | 2.748973 | 2.830353 | 3.241517 | 2.657959 | 2.247117 | 2.164293 | 2.359059 | 2.657804 | 3.155508 |

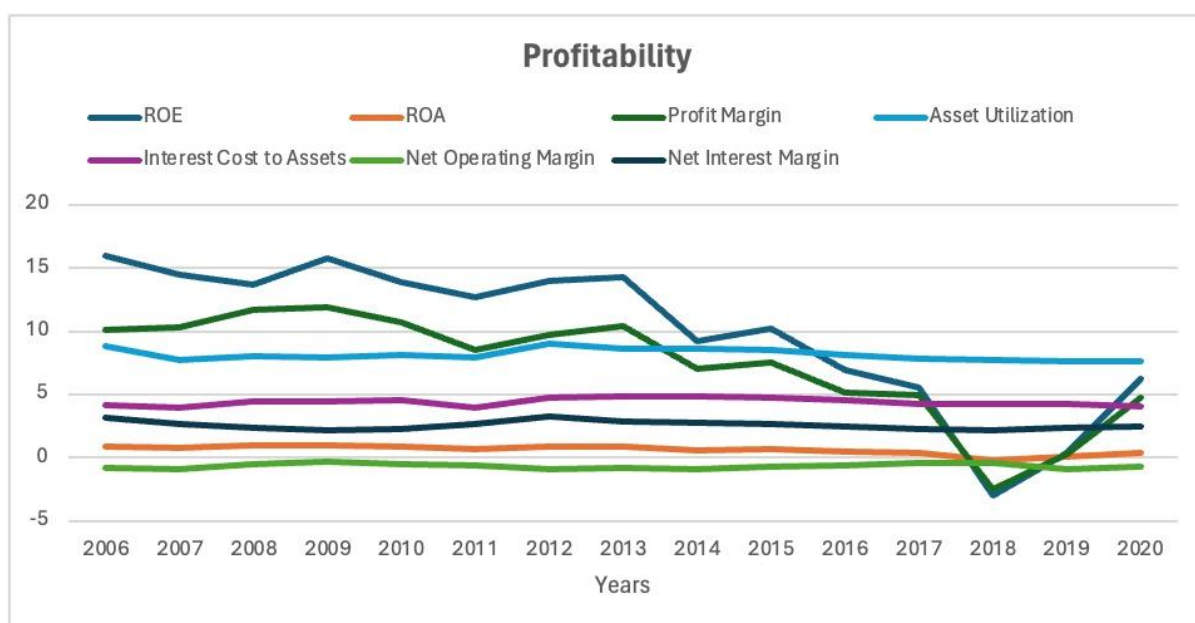


Figure 4 Profitability

The State Bank of India's (SBI) profitability ratios from 2006 to 2020 are shown in the chart. Key performance indicators including asset utilization, profit margin, net interest margin, return on equity (ROE), return on assets (ROA), and interest cost to assets are included. Based on the patterns in the chart, the following is a thorough analysis of each ratio:

**1. Trend for Return on Equity (ROE):** The profitability as a percentage of shareholders' equity, or ROE, starts out strongly between 12% and 15% and then gradually declines after 2013. Around 2017, there is the biggest decline, which is followed by a rebound in 2019.

**Evaluation:** The drop in ROE could be the result of growing costs or larger provisions for bad loans, which would reduce shareholder profitability.

**2. Trend for Return on Assets (ROA):** The ROA remains very low throughout the years, fluctuating between 0% and 1%, indicating that SBI continually operates with lower returns on its total assets.

**Analysis:** Given their enormous volume and narrow profit margins, big banks such as SBI tend to have low ROA. But the drop after 2017 and the modest rebound by 2020 point to a time of lower asset profitability, which could be brought on by bad loans, inaccurate provisioning, or problems with asset management.

**3. Profit Margin Trend:** The profit margin began at roughly 10% and peaked in 2010, after which it declined steadily until 2017. This pattern is comparable to that of ROE. There is a noticeable decrease in 2017, and then it stabilizes and starts to rebound in 2020.

**Analysis:** Based on the profit margin, it appears that SBI's capacity to turn a profit declined dramatically in the following years, especially after 2015. This could be a result of decreased lending margins, regulatory difficulties, or a rise in non-performing assets (NPAs). The rebound suggests enhanced cost control or higher operational effectiveness after 2018.

**4. Trend for Net Interest Margin (NIM):** The NIM varies between 2.5% and 3%, showing a minor decline from 2015 to 2018 before levelling off in the 2019–2020 range.

**Evaluation:** A falling NIM indicates that SBI's capacity to make money from its lending and investment operations was probably being impacted by heightened competition or a low-interest rate environment. The stabilization following 2019 can be a sign of better asset management or higher loan rates.

**5. Net Operating Margin Trend:** The net operating margin peaked about 2010 and then declined until 2018, following a pattern close to that of ROE and profit margin. It appears to be improving as we approach 2020.

**Analysis:** Over time, SBI may have battled with higher operating costs or operational inefficiencies, as evidenced by the declining operating margin. The improvement is consistent with the overall recovery in other profitability ratios, suggesting cost-cutting initiatives or more effective operations in subsequent years.

**6. Asset Utilization Trend:** Up until 2018, asset utilization had a rather flat trend of 0.5%. After that, it fell sharply, peaked in 2019, and then began to gradually rise again in 2020.

**Analysis:** SBI's asset utilizations ratio shows how well the company uses its resources to produce income. The notable reduction observed after 2017 may be the result of either a large decrease in activities that generate revenue or an increase in inefficient assets (such non-performing loans). The little improvement in 2020 can be a sign of asset quality improvement initiatives.

**7. Cost of Interest on Assets:** Generally, with just little variations, this ratio stays largely constant over the course of time. It implies that SBI regularly kept its interest costs within a constrained range as compared to its total assets.

**Analysis:** SBI's ability to control its borrowing costs in relation to its assets is demonstrated by a constant interest cost to asset ratio. Any growth after 2017 could, however, point to increased funding costs or the need to raise interest rates to draw in more deposits.

## 4.2 Risk:

| Risk              | 2020     | 2019     | 2018     | 2017     | 2016     | 2015     | 2014     | 2013     | 2012     | 2011     | 2010     | 2009     | 2008     | 2007     | 2006     |
|-------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Equity Multiplier | 17.03132 | 16.66222 | 15.76587 | 14.37157 | 16.3412  | 15.94603 | 15.15653 | 15.83892 | 15.90828 | 18.83075 | 15.97311 | 16.64315 | 14.71522 | 18.10196 | 17.87105 |
| Equity Ratio      | 0.058715 | 0.060016 | 0.063428 | 0.069582 | 0.061195 | 0.062712 | 0.065978 | 0.063136 | 0.06286  | 0.053105 | 0.062605 | 0.060085 | 0.067957 | 0.055243 | 0.055956 |
| Capital Adequacy  | 14.92857 | 18.24772 | 18.03586 | 20.3117  | 21.45277 | 17.70645 | 18.24318 | 18.55919 | 17.11076 | 16.83746 | 17.54868 | 16.58575 | 15.4852  | 13.87358 | 13.04751 |
| Provision Ratio   | 66.17673 | 66.61944 | 62.57683 | 63.7512  | 68.0394  | 68.46589 | 69.91483 | 69.81871 | 68.93887 | 65.4142  | 63.32051 | 60.16354 | 63.91801 | 64.00477 | 57.5233  |
| Net NPA to Asset  | 2.415536 | 3.210522 | 4.555148 | 2.660018 | 2.751322 | 1.478928 | 2.252814 | 2.184421 | 1.294609 | 0.490504 | 0.824125 | 0.783014 | 0.895126 | 0.801209 | 0.705059 |
| Net NPA to Equity | 41.13978 | 53.49442 | 71.81586 | 38.22864 | 44.9599  | 23.58303 | 34.14484 | 34.59888 | 20.595   | 9.238552 | 13.16383 | 13.03182 | 13.17198 | 14.50345 | 12.60014 |

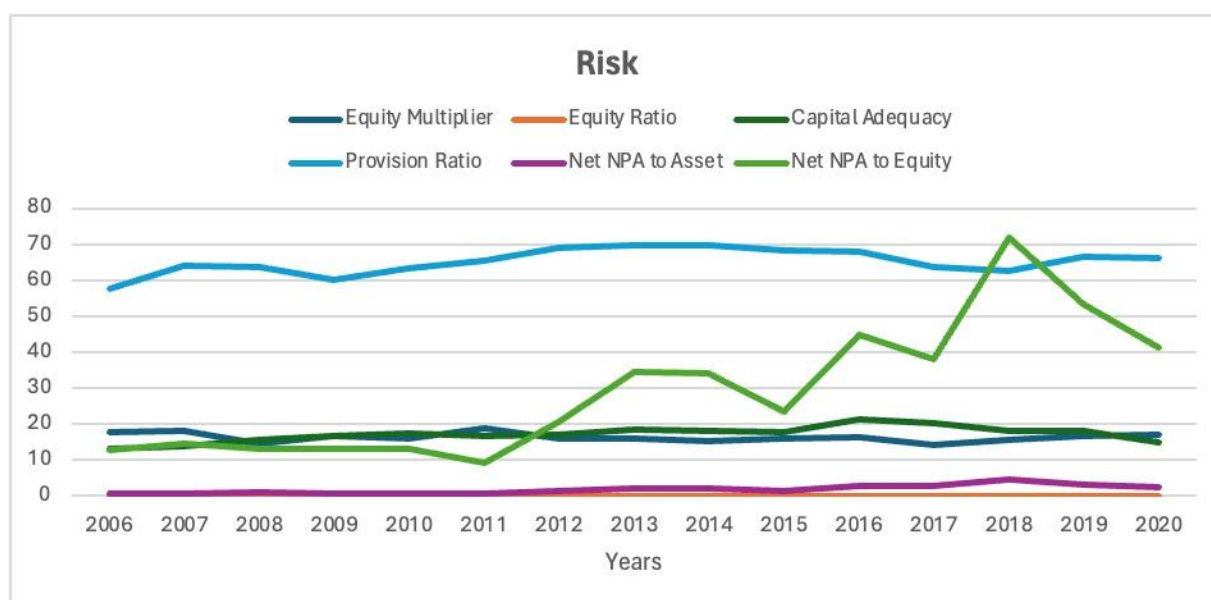


Figure 5 Risk

The provision ratio starts from 60 in 2006 and reaches up to 68% in 2020 compared to the net NPA to equity ratio starts from 11% which drastically increases to 40%. The equity

multiplier and the capital adequacy ratio are almost at the same line with both intersecting each other starting from 19% in 2006 to approximately 17-18% in the year 2020.

### 4.3 Liquidity ratio:

| Liquidity                    | 2020     | 2019     | 2018     | 2017     | 2016     | 2015     | 2014     | 2013     | 2012     | 2011     | 2010     | 2009     | 2008     | 2007     | 2006     |
|------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Demand Deposit               | 5.753301 | 5.593047 | 5.504704 | 5.632779 | 5.930013 | 6.082395 | 6.316139 | 7.194449 | 7.371689 | 10.72088 | 11.6364  | 11.48381 | 13.60082 | 14.47282 | 13.76349 |
| SLR-Total Investment         | 0.679352 | 0.713708 | 0.634953 | 0.706528 | 0.870307 | 0.91402  | 0.98898  | 1.026651 | 0.983894 | 0.993559 | 0.890354 | 0.838777 | 0.9138   | 0.949664 | 0.759884 |
| Cash to Demand Deposit Ratio | 0.973953 | 0.971233 | 0.85261  | 0.949405 | 1.01739  | 1.057623 | 1.06116  | 0.751357 | 0.742415 | 0.839063 | 0.565563 | 0.807381 | 0.6671   | 0.58199  | 0.579423 |
| Cash to Total Deposit Ratio  | 6.830356 | 6.867962 | 5.991262 | 7.077115 | 8.218435 | 8.355603 | 8.617111 | 7.039185 | 7.003418 | 11.78684 | 8.62144  | 12.05007 | 12.18169 | 10.95745 | 10.3667  |

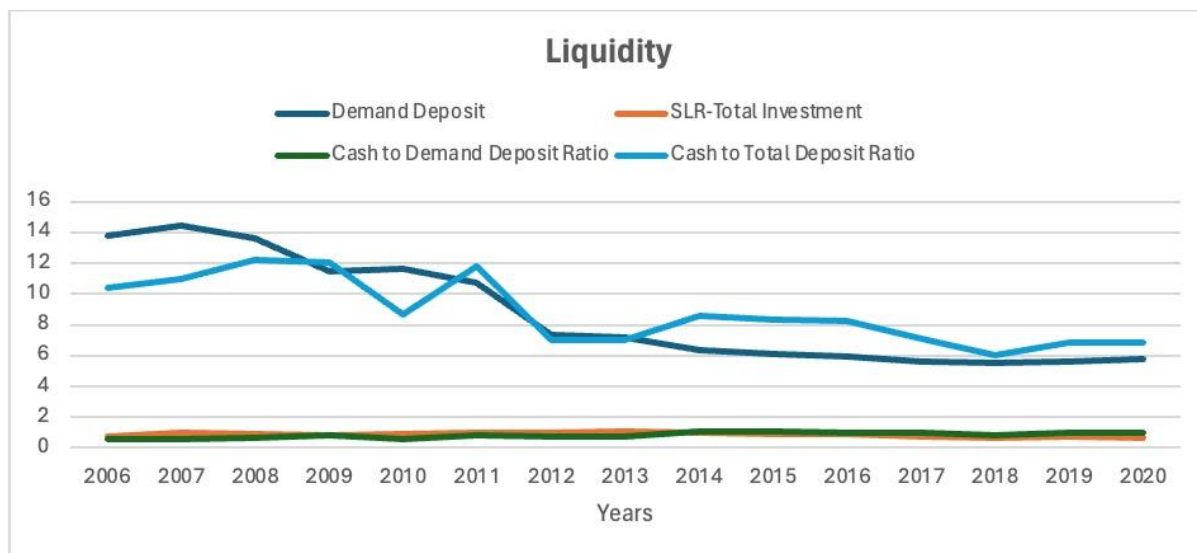


Figure 6 Liquidity

The demand deposit ratio generally declines over the period because the customer preference towards the deposit product shifts. The SLR remains stable indicating that SBI has maintained compliance with regulatory requirements. The cash to demand deposit ratio has experienced fluctuations with peak period of higher liquidity. The cash to total deposit ratio shows a similar pattern to that of the demand deposit ratio, which fluctuates drastically and remains stable for the year 2020.

#### 4.4 Efficiency and Expense Control Ratio:

| Efficiency and Expense Control Ratio | 2020     | 2019     | 2018     | 2017     | 2016     | 2015     | 2014     | 2013     | 2012     | 2011     | 2010     | 2009     | 2008     | 2007     | 2006     |
|--------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Operating Efficiency                 | 1.90246  | 1.893218 | 1.735101 | 1.717419 | 1.772228 | 1.858027 | 1.992798 | 1.869762 | 1.951974 | 1.880751 | 1.928841 | 1.622582 | 1.747491 | 2.086876 | 2.373362 |
| Cost of Funds                        | 4.477683 | 4.66207  | 4.746498 | 4.811054 | 5.199611 | 5.464922 | 5.519268 | 5.490529 | 5.401291 | 4.638621 | 5.216738 | 5.194732 | 5.419687 | 4.668139 | 4.964957 |
| Income Productive per Employee       | 1.209866 | 1.09994  | 1.010525 | 0.784054 | 0.897708 | 0.813167 | 0.712547 | 0.60015  | 0.52041  | 0.442844 | 0.378466 | 0.375359 | 0.275148 | 0.240569 | 0.229888 |
| Efficiency Ratio                     | 0.898901 | 0.993109 | 1.054811 | 0.892273 | 0.882989 | 0.831146 | 0.839446 | 0.766343 | 0.796898 | 0.829072 | 0.762781 | 0.728243 | 0.738331 | 0.791907 | 0.808608 |

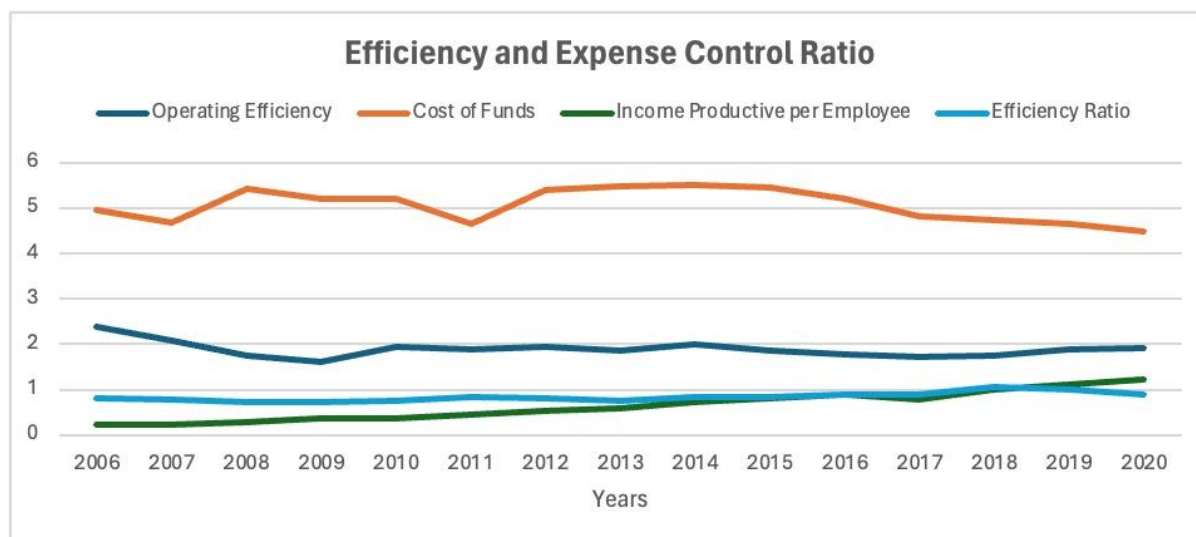


Figure 7 Efficiency and Expense Control ratio

The operating efficiency ratio fluctuated over the time and showed improvement and others showed decline. A higher operating efficiency ratio indicates SBI was able to generate more income per unit of operating expense. The cost of funds experienced some fluctuations and remained relatively stable. The lower cost of funds implies that SBI was able to finance at cheaper rate which can also help in improving the profitability of SBI. The income productivity per employee suggests that SBI productivity per employee was declining and showed a downward trend. The efficiency ratio fluctuated but remained stable, which indicates that SBI overall efficiency was consistent.



## 5. YES Bank:

### 5.1 Profitability:

| Profitability           | 2020     | 2019     | 2018     | 2017     | 2016     | 2015     | 2014     | 2013     | 2012     | 2011     | 2010     | 2009     | 2008     | 2007     | 2006     |
|-------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| ROE                     | -75.5676 | 6.394091 | 16.4008  | 15.0997  | 18.41967 | 17.16922 | 22.71608 | 22.39591 | 20.89105 | 19.16509 | 15.46307 | 18.70699 | 15.16575 | 11.98961 | 9.660425 |
| ROA                     | -6.36785 | 0.451723 | 1.352096 | 1.548451 | 1.536605 | 1.472685 | 1.483987 | 1.312438 | 1.32698  | 1.232291 | 1.313102 | 1.326775 | 1.177832 | 0.849872 | 1.3291   |
| Profit Margin           | -138.473 | 37.47759 | 80.87095 | 80.11285 | 93.63233 | 97.99196 | 93.97081 | 103.4394 | 113.9861 | 116.6648 | 83.00829 | 69.54483 | 55.45931 | 48.49878 | 56.96732 |
| Asset Utilization       | 4.598625 | 1.205314 | 1.671918 | 1.932837 | 1.641106 | 1.502863 | 1.5792   | 1.268799 | 1.16416  | 1.056266 | 1.581892 | 1.907798 | 2.123776 | 1.752357 | 2.333091 |
| Interest Cost to Assets | 7.47066  | 5.203349 | 4.010414 | 4.94157  | 5.425714 | 5.936803 | 6.684256 | 6.130127 | 6.372398 | 4.736418 | 4.347575 | 6.515651 | 5.735983 | 3.748895 | 2.515648 |
| Net Operating Margin    | 1.989653 | -0.4396  | 0.003538 | 0.0187   | -0.15988 | -0.17497 | -0.02595 | -0.0778  | -0.10243 | -0.09582 | 0.207184 | 0.080153 | 0.114843 | 0.009656 | -2.06893 |
| Net Interest Margin     | 2.639457 | 2.575724 | 2.47629  | 2.695671 | 2.763299 | 2.561377 | 2.491621 | 2.238847 | 2.194393 | 2.11319  | 2.165746 | 2.223937 | 1.94659  | 1.543239 | 2.116191 |

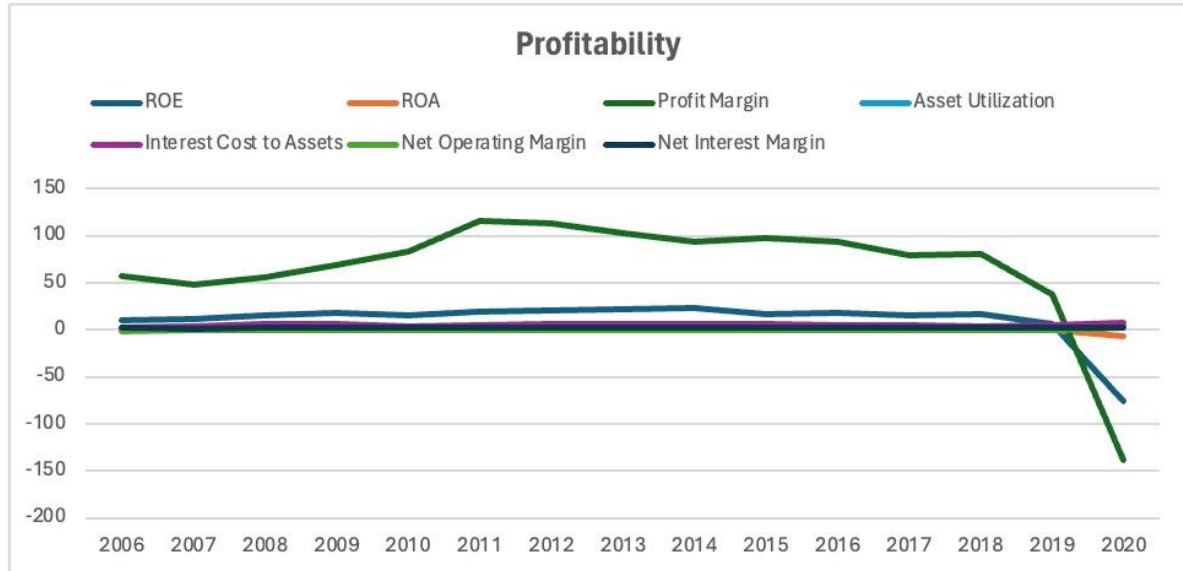


Figure 8 Profitability of YES Bank

The asset utilization, the interest cost to asset ratio, the net interest margin, the return on equity, the return on asset remained stagnant approximately to 0-30%. But the profit margin fluctuated from 2006 and reached at a peak in 2011, and remained stable until 2016, but the profit margin drastically declines and reach at a point zero and turned negative in the year 2020 because of the collabs of YES bank in the year 2020. The return on equity also turned negative in the year 2020.

## 5.2 Risk:

| Risk              | 2020     | 2019     | 2018     | 2017     | 2016     | 2015     | 2014     | 2013     | 2012     | 2011     | 2010     | 2009     | 2008     | 2007     | 2006     |
|-------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Equity Multiplier | 11.86705 | 14.1549  | 12.12991 | 9.751489 | 11.98725 | 11.65844 | 15.30746 | 17.06435 | 15.7433  | 15.5524  | 11.77599 | 14.0996  | 12.87599 | 14.10755 | 7.268397 |
| Equity Ratio      | 0.084267 | 0.070647 | 0.082441 | 0.102548 | 0.083422 | 0.085775 | 0.065328 | 0.058602 | 0.063519 | 0.064299 | 0.084919 | 0.070924 | 0.077664 | 0.070884 | 0.137582 |
| Capital Adequacy  | 54.54822 | 38.32364 | 35.08515 | 31.1357  | 29.02493 | 29.54175 | 27.65326 | 28.19339 | 26.95721 | 18.93326 | 23.33355 | 25.54756 | 15.11196 | 16.9859  | 26.51813 |
| Provision Ratio   | 79.32454 | 69.21731 | 69.92574 | 67.32781 | 65.19035 | 62.4701  | 56.76269 | 52.30133 | 57.18886 | 61.94138 | 64.27224 | 59.95405 | 59.82643 | 60.04094 | 61.55021 |
| Net NPA to Asset  | 1.755847 | 0.38546  | -0.287   | -0.22732 | -0.61361 | -0.68357 | -0.82123 | -0.75342 | -0.65103 | -0.64861 | -0.89436 | -0.60645 | -0.81883 |          |          |
| Net NPA to Equity | 20.83672 | 5.456144 | -3.48128 | -2.21669 | -7.35548 | -7.9694  | -12.5709 | -12.8567 | -10.2494 | -10.0874 | -10.532  | -8.55077 | -10.5432 |          |          |

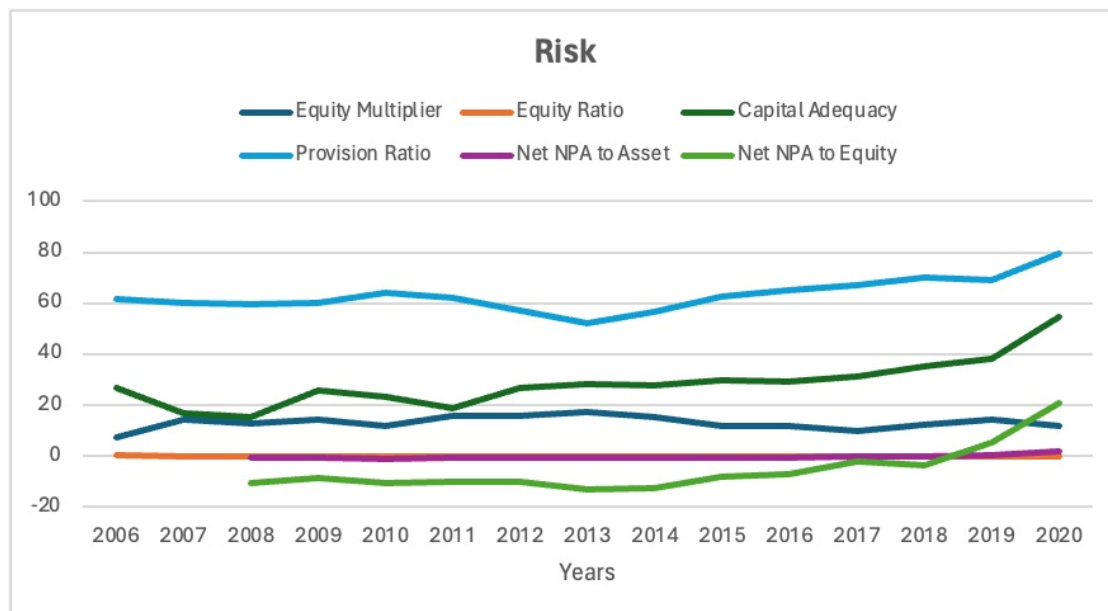


Figure 9 Risk of YES Bank

The equity ratio and the net NPA to asset ratio remained stable throughout the year 2006-2020. The net NPA to equity ratio was negative in the year 2008, but slowly and gradually it started inclining in the year 2018 until 2020. The provision ratio was 60% in 2006 which declined in 2013 and again it inclined in 2020 with a proximate of 80%. The equity multiplier and the capital adequacy ratio also fluctuated since 2006, but the capital adequacy ratio steadily increased in the year 2020 with approximate 58%.

### 5.3 Liquidity:

| Liquidity                    | 2020     | 2019     | 2018     | 2017     | 2016     | 2015     | 2014     | 2013     | 2012     | 2011     | 2010     | 2009     | 2008     | 2007     | 2006     |
|------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Demand Deposit               | 3.684375 | 7.494781 | 9.225838 | 8.875582 | 6.610705 | 6.241774 | 6.43683  | 6.725124 | 6.63948  | 6.666718 | 6.671235 | 5.326217 | 5.783214 | 3.745558 | 7.229381 |
| SLR-Total Investment         | 16.19615 | 7.709515 | 9.849226 | 10.81699 | 10.25818 | 10.18627 | 9.631285 | 8.382079 | 11.06624 | 15.59824 | 25.79382 | 32.52256 | 33.99613 | 46.09033 | 91.47406 |
| Cash to Demand Deposit Ratio | 0.83477  | 0.804941 | 0.807783 | 0.905687 | 0.593956 | 0.785963 | 0.706655 | 0.535331 | 0.726583 | 0.828271 | 1.025908 | 1.42446  | 1.59671  | 3.026669 | 0.680153 |
| Cash to Total Deposit Ratio  | 7.526045 | 10.09388 | 11.59966 | 12.09989 | 5.808307 | 7.326774 | 6.68362  | 5.328777 | 7.2262   | 7.092625 | 9.291709 | 10.74546 | 11.81465 | 15.31253 | 7.032634 |

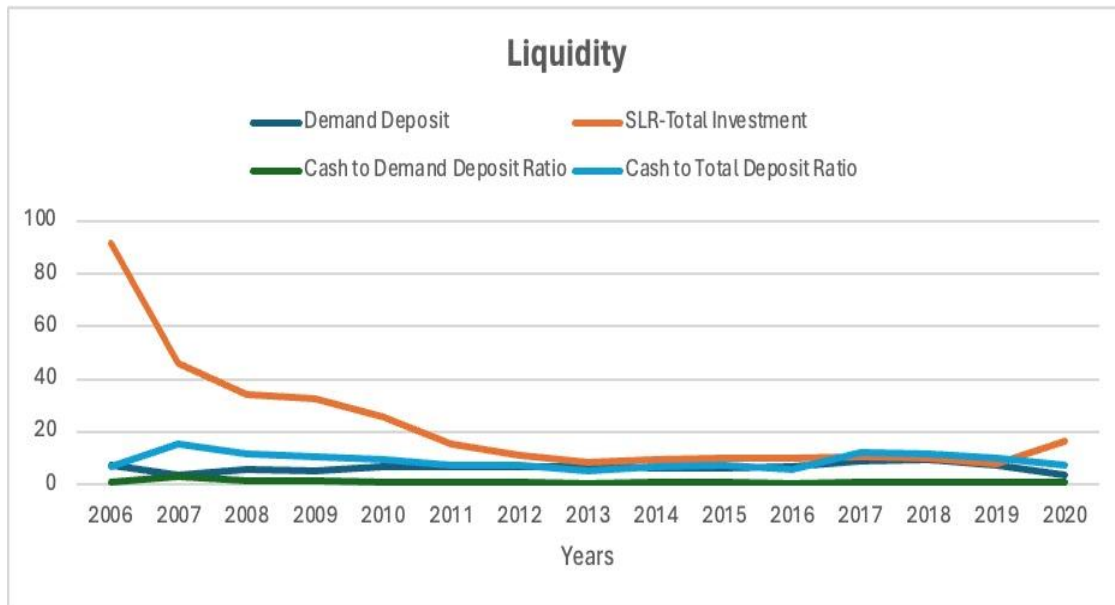


Figure 10 Liquidity of YES Bank

The demand deposit ratio, cash to demand deposit, and cash to total deposit ratio remained stable or negligible fluctuations from the year 2006 to 2020 for YES bank which affect the liquidity creation of the YES bank. The SLR to investment ratio was usually high in 2006, but after 2009, it drastically declined and became almost 16% in 2020.

## 5.4 Efficiency and Expense Control Ratio:

| Efficiency and Expense Control Ratio | 2020     | 2019     | 2018     | 2017     | 2016     | 2015     | 2014     | 2013     | 2012     | 2011     | 2010     | 2009     | 2008     | 2007     | 2006     |
|--------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Operating Efficiency                 | 2.609972 | 1.644918 | 1.66838  | 1.914137 | 1.800986 | 1.677829 | 1.605155 | 1.346601 | 1.266588 | 1.152084 | 1.374707 | 1.827645 | 2.008934 | 1.742701 | 2.068934 |
| Cost of Funds                        | 8.788949 | 5.896933 | 4.548052 | 5.85591  | 6.25388  | 6.886225 | 7.606924 | 6.913252 | 7.410923 | 5.310328 | 5.013867 | 7.509075 | 6.831359 | 4.580431 | 3.102543 |
| Income Productive per Employee       | 51.23598 | 18.16376 | 22.56556 | 20.71266 | 11.96223 | 11.86755 | 13.8055  | 12.23289 | 10.41542 | 9.510876 | 12.78907 | 13.09562 | 12.31614 | 5.668178 | 3.659312 |
| Efficiency Ratio                     | 67.42644 | 27.21734 | 16.69221 | 16.0549  | 14.62239 | 13.53962 | 13.25889 | 11.9171  | 9.367767 | 6.695729 | 5.621601 | 4.866317 | 5.442601 | 3.914875 | 1.726363 |

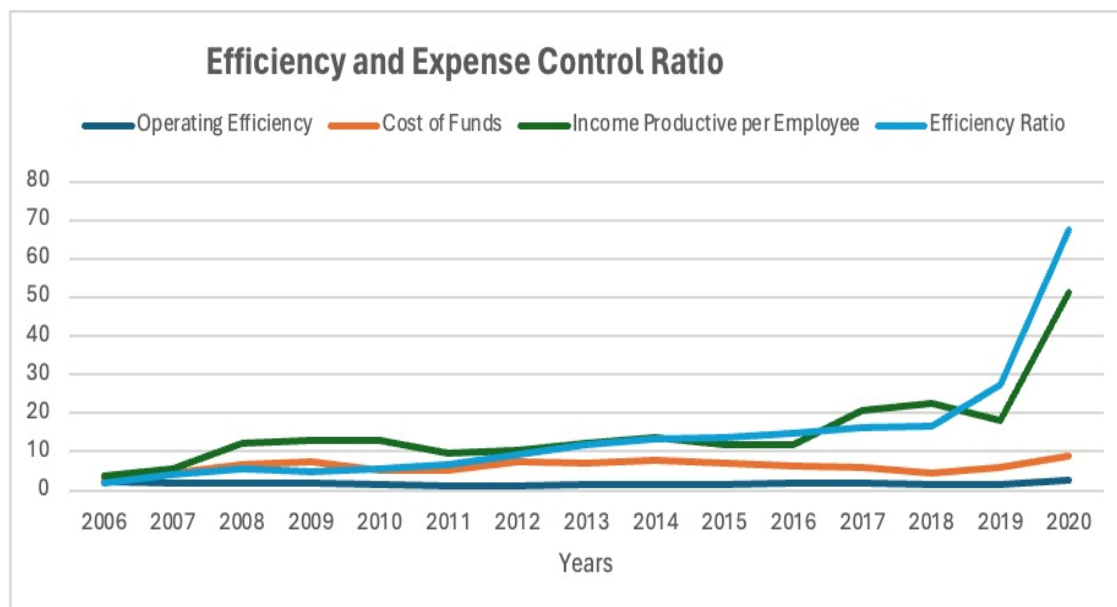


Figure 11 Efficiency and Expense ratio for YES Bank

The operating efficiency ratio, cost of funds, income productive per employee, and the efficiency ratio effects the performance indicators of YES bank. Initially the cost of funds and operating efficiency remains stable from 2006-2020. The income productive per employee and efficiency ratio gradually increased from the year 2016 and reached a peak of 70% approx. in the year 2020.

## 6. Comparison of YES Bank and SBI based on their benchmark:

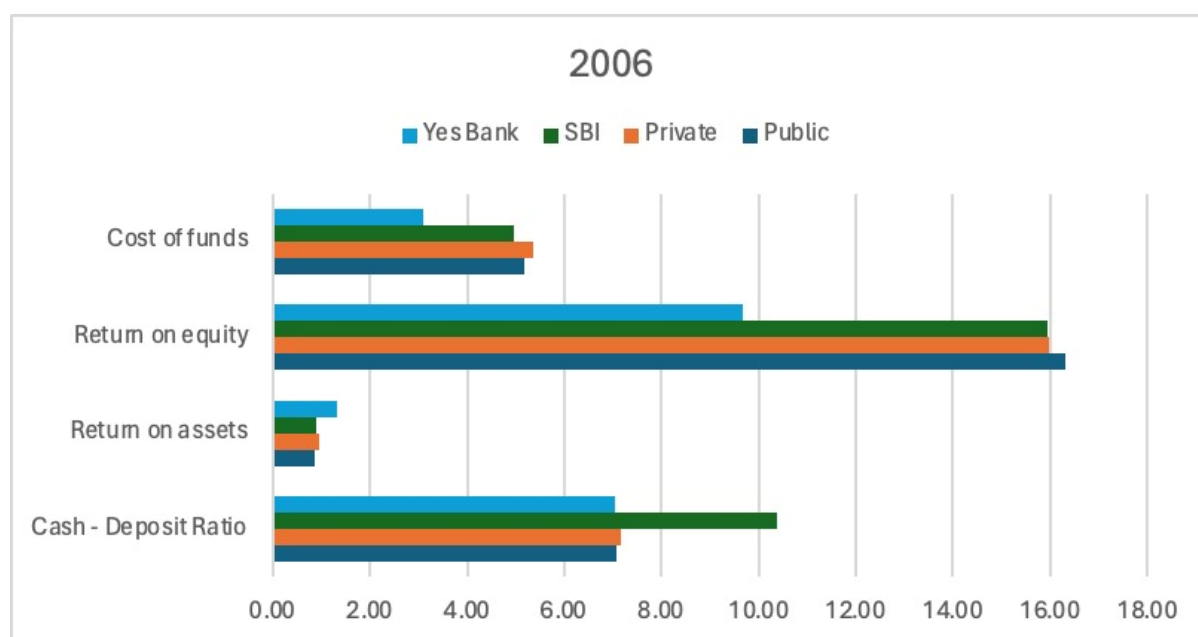


Figure 12 Comparative Analysis of Banks based on their benchmark

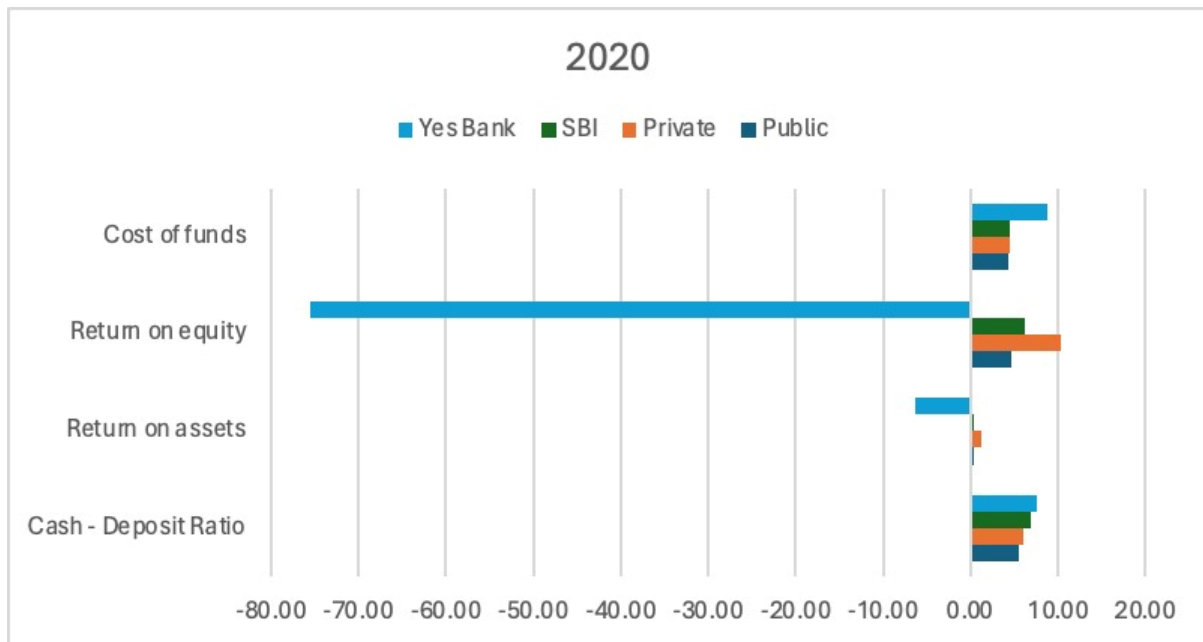
Cost of Funds: SBI most certainly had a competitive advantage in terms of profitability due to its reduced cost of funds.

Return on Equity: YES Bank's higher ROE indicates that, maybe as a result of its more adaptable business practices or risk-taking tactics, it was able to produce larger returns for its investors.

Return on Assets: SBI's overall profitability may have been influenced by its lower cost of funding, even if Yes Bank had a little higher ROA.

Cash-Deposit Ratio: The disparity in the two banks' cash-deposit ratios may be due to differences in their approaches to managing liquidity or their respective levels of risk appetite.

As a whole: In comparison to SBI, Yes Bank seems to have had a higher ROE and ROA based on the scant data that is available for 2006. SBI may have had a competitive advantage in terms of overall profitability, nonetheless, due to its cheaper cost of financing. Further analysis with more data points and contextual information would be essential to establish more definitive results.



*Figure 13 Comparative study of banks based on benchmark*

SBI seems to have had a major advantage in terms of cost of funding and overall profitability, based on the figures for 2020. YES Bank had several difficulties, such as low returns and cash flow problems.

Nonetheless, Yes Bank's better ROE can mean that, thanks to restructuring or government assistance, some value was recovered for the company's stockholders. It's crucial to remember that the COVID-19 pandemic made 2020 very difficult for the banking industry, thus these findings might not be indicative of typical business circumstances.